

Data Handlers

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BES Data Handlers

- Data Handlers are specialized BES Modules that know how to read data formats and create DAP2 objects.
- Within the BES configuration you specify the modules that you want to load into the BES.
- The module can add named handlers to the different pieces of the BES.
- The module can extend already existing modules, for example extending the DAP2 module.
- The module can modify already existing modules, such as changing the way certain responses are built.

Getting the Data Handlers

There are two ways to ways to get data handler modules

1. Download and install the binaries
2. Download and build the source tarball



Download

ODC
Matlab - loaddap
Matlab - toolbox
IDL-Client
libdap
libnc-dap
Java library
Ocapi
Hyrax
CGI Server

FTP

Nightly Builds

SVN

Libraries our software uses

Software from other groups

http://www.opendap.org/download/data_handlers.html

Handlers

This page lists all of the data handlers we provide, in any form, for either the [CGI Server](#) or for [Hyrax](#).

There are three broad classifications of handlers listed here:

1. Handlers for which we provide full support: We ensure these work with the latest data server and we provide binary builds.
2. Handlers for which we provide source releases: We often test these against our current server, but not always. We don't, as a rule, provide binary builds for these handlers.
3. Handlers for which we do not release on a schedule: These are stored in our source code management system (which uses 'subversion') which is open for read-access to all and write-access by request.

If you've written, or are in the process of writing a handler, and you would like us to distribute it, let [someone at OPeNDAP know](#).

Handlers with full support

HDF4

This handler reads data stored in HDF 4 and HDF-EOS files. To run this you will also need the HDF 4 libraries on your system.

NetCDF 3

This handler reads data stored in netCDF 3 files. It looks for COARDS-compliant files and, if found, promotes arrays to DAP Grid variables so that software that's not COARDS-compliant can make use of bound dimensions.

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libnc-dap
Java library
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Hyrax
CGI Server

FTP**Nightly Builds****SVN****Libraries our software
uses****Software from other
groups**

netCDF handler

This is the netCDF file module for the [CGI Server](#) and for [Hyrax](#).

Note: This page lists our 'full releases' of the handler. Sometimes we make source-only releases or internal releases and don't bother to include them here. That is why the version numbers you see here have 'gaps.'

netCDF handler 3.7.6 (13 March 2007)

Binaries

 **Linux (x86_64)**
[Fedora Core 5, RPMs](#) (gcc 4.1)

 **Linux (i386)**
[Fedora Core 4, RPMs](#) (gcc 4.0)

 **Apple OS/X (10.4)**
[Intel DMG](#)

 **SPARC/Solaris**
Solaris 5.8

Source code

Version 3.7.6

[tar.gz](#). Required: [bes 3.4.1](#) (for Hyrax -this is required to build the BES modules).

Downloading Source Tarball

What we're going to do is download and build from a source tarball.

- On the handler page, under Source code, click on tar.gz
- Save it to wherever you are going to do your build.
- For this demo, we've installed the source tarball in: `/root/src/netcdf_handler-3.7.6`

```
root@slax:~# ls
Desktop/  Set\ IP\ address  src/
root@slax:~#
root@slax:~# cd src
root@slax:~/src# ls
bes-3.5.1/  dap-server-3.7.4/  libdap-3.7.7/  netcdf-3.6.2/
bes.log    freeform_handler-3.7.5/  libnc-dap-3.7.0/  netcdf_handler-3.7.6/
root@slax:~/src#
root@slax:~/src# cd netcdf_handler-3.7.6/
root@slax:~/src/netcdf_handler-3.7.6# ls
COPYING      NCFloat32.h      NCRequestHandler.h  NCUIInt32.h      configure*
COPYRIGHT    NCFloat64.cc     NCSequence.cc       NCUrl.cc         configure-nc-data.sh.in
ChangeLog    NCFloat64.h      NCSequence.h        NCUrl.h          configure.ac
INSTALL      NCGrid.cc        NCStr.cc            NEWS             data/
Makefile     NCGrid.h         NCStr.h             README          libtool*
Makefile.am  NCInt16.cc       NCStructure.cc      aclocal.m4      nc_handler.cc
Makefile.in  NCInt16.h        NCStructure.h       conf/           ncdas.cc
NCArray.cc   NCInt32.cc       NCTypeFactory.cc   config.h         ncdds.cc
NCArray.h    NCInt32.h        NCTypeFactory.h    config.h.in     netcdf_handler.spec
NCByte.cc    NCModule.cc      NCUIInt16.cc       config.log       stamp-h1
NCByte.h     NCModule.h       NCUIInt16.h        config.status*
NCFloat32.cc NCRequestHandler.cc NCUIInt32.cc       config_nc.h
```

Building the Data Handler

1. `cd src/netcdf_handler-3.7.6`
2. `./configure`
3. `make`
4. `make install`
5. `export LD_LIBRARY_PATH=/usr/local/lib`

Configuring the Data Handler

Edit the bes.conf file /usr/local/etc/bes/
bes.conf

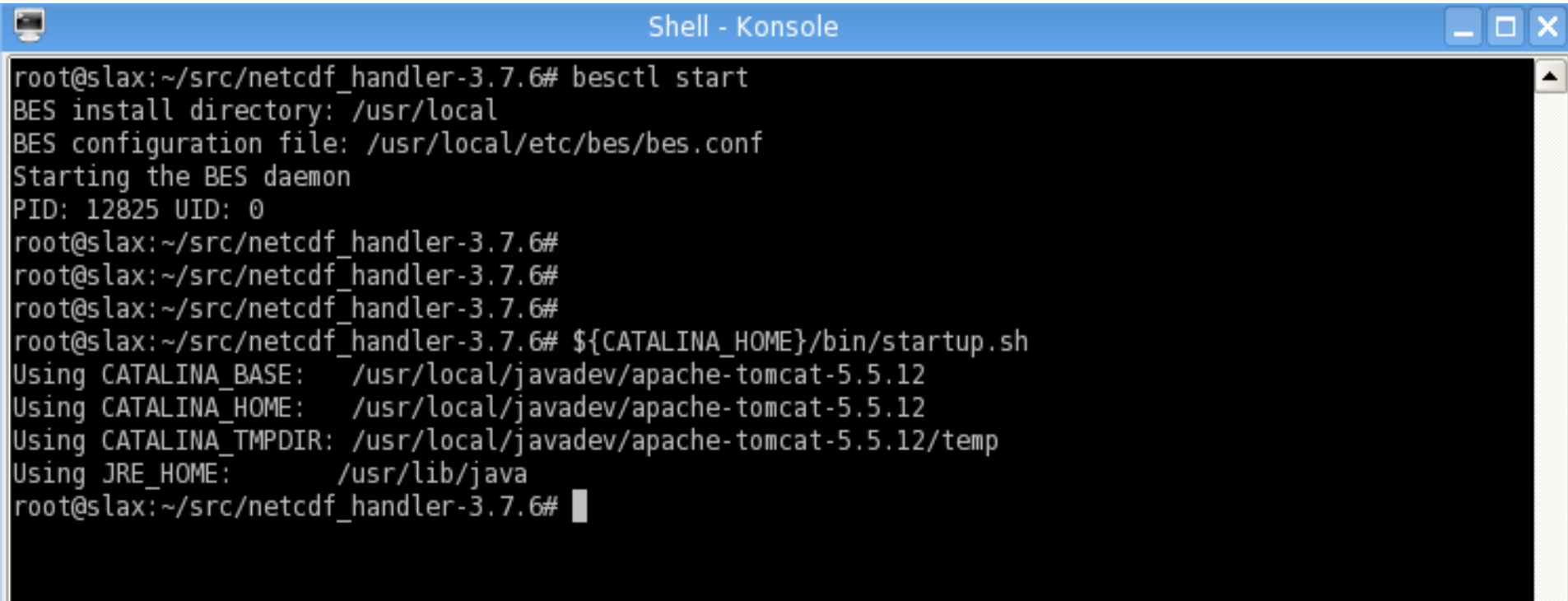
- Add to the BES.modules list
BES.modules=dap,cmd,usage,ascii,www,nc,ff
- Add the module
BES.module.nc=/usr/local/lib/bes/libnc_module.so
- Update the data root directory
BES.Catalog.catalog.RootDirectory=/usr/local/share/hyrax
- Add to BES.Catalog.catalog.TypeMatch
nc:.*\.nc(\.bz2|\.gz|\.Z)?\$;

Or just run configure-nc-data.sh installed
in /usr/local/bin.

Restart Hyrax

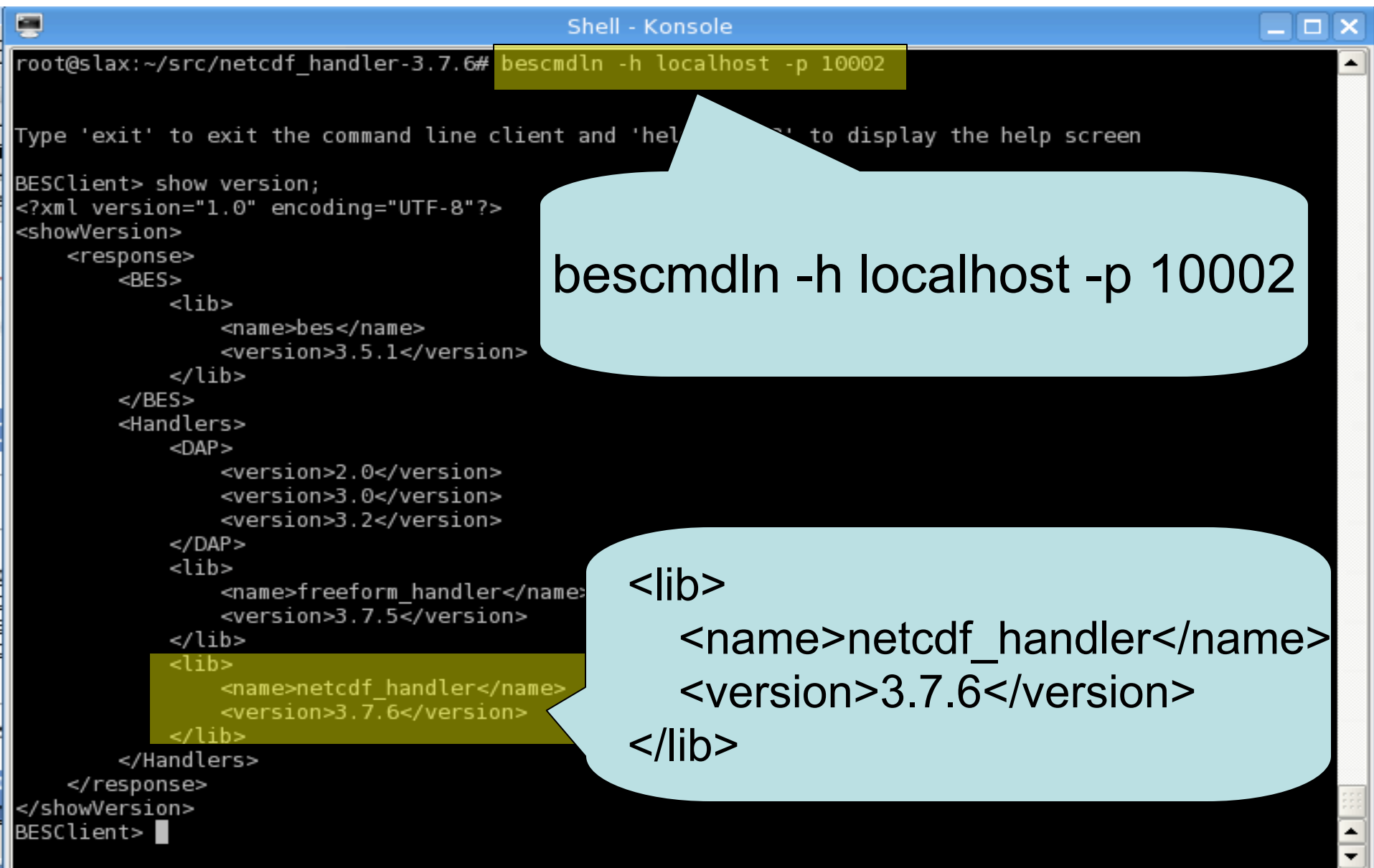
- If you have tomcat and the BES running
 - besctl stop
 - \${CATALINA_HOME}/bin/shutdown.sh
- Now restart the BES
 - besctl start
- And restart tomcat
 - \${CATALINA_HOME}/bin/startup.sh

... screen shot



```
root@slax:~/src/netcdf_handler-3.7.6# besctl start
BES install directory: /usr/local
BES configuration file: /usr/local/etc/bes/bes.conf
Starting the BES daemon
PID: 12825 UID: 0
root@slax:~/src/netcdf_handler-3.7.6#
root@slax:~/src/netcdf_handler-3.7.6#
root@slax:~/src/netcdf_handler-3.7.6#
root@slax:~/src/netcdf_handler-3.7.6# ${CATALINA_HOME}/bin/startup.sh
Using CATALINA_BASE:   /usr/local/javadev/apache-tomcat-5.5.12
Using CATALINA_HOME:   /usr/local/javadev/apache-tomcat-5.5.12
Using CATALINA_TMPDIR: /usr/local/javadev/apache-tomcat-5.5.12/temp
Using JRE_HOME:        /usr/lib/java
root@slax:~/src/netcdf_handler-3.7.6#
```

Check that it was successful



The image shows a terminal window titled "Shell - Konsole". The prompt is "root@slax:~/src/netcdf_handler-3.7.6#". The command "bescmdln -h localhost -p 10002" is entered and highlighted in a yellow box. Below the command, a light blue speech bubble contains the text "bescmdln -h localhost -p 10002". The terminal output shows a message: "Type 'exit' to exit the command line client and 'help' to display the help screen". Then, the command "BESClient> show version;" is entered. The output is an XML document. A light blue speech bubble contains a snippet of this XML: "<lib>\n <name>netcdf_handler</name>\n <version>3.7.6</version>\n</lib>". In the terminal, the corresponding XML snippet is highlighted in a yellow box.

```
root@slax:~/src/netcdf_handler-3.7.6# bescmdln -h localhost -p 10002

Type 'exit' to exit the command line client and 'help' to display the help screen

BESClient> show version;
<?xml version="1.0" encoding="UTF-8"?>
<showVersion>
  <response>
    <BES>
      <lib>
        <name>bes</name>
        <version>3.5.1</version>
      </lib>
    </BES>
    <Handlers>
      <DAP>
        <version>2.0</version>
        <version>3.0</version>
        <version>3.2</version>
      </DAP>
      <lib>
        <name>freeform_handler</name>
        <version>3.7.5</version>
      </lib>
      <lib>
        <name>netcdf_handler</name>
        <version>3.7.6</version>
      </lib>
    </Handlers>
  </response>
</showVersion>
BESClient> 
```

... or use the OLFS

OPeNDAP Hyrax: Contents of /data/nc/ - Mozilla Firefox

File Edit View History Bookmarks Tools Help

http://localhost:8080/opendap/data/nc/contents.html

Getting Started Latest Headlines

OPeNDAP

Contents of /data/nc/

Name	Last modified	Size	Response Links
Parent Directory	-	-	
coads_climatology.nc	2007-10-01 05:56:37	2 Mbytes	ddx dds das info html
fnoc1.das	2007-10-01 05:56:37	162 bytes	- - - -
fnoc1.nc	2007-10-01 05:56:37	23 Kbytes	ddx dds das info html

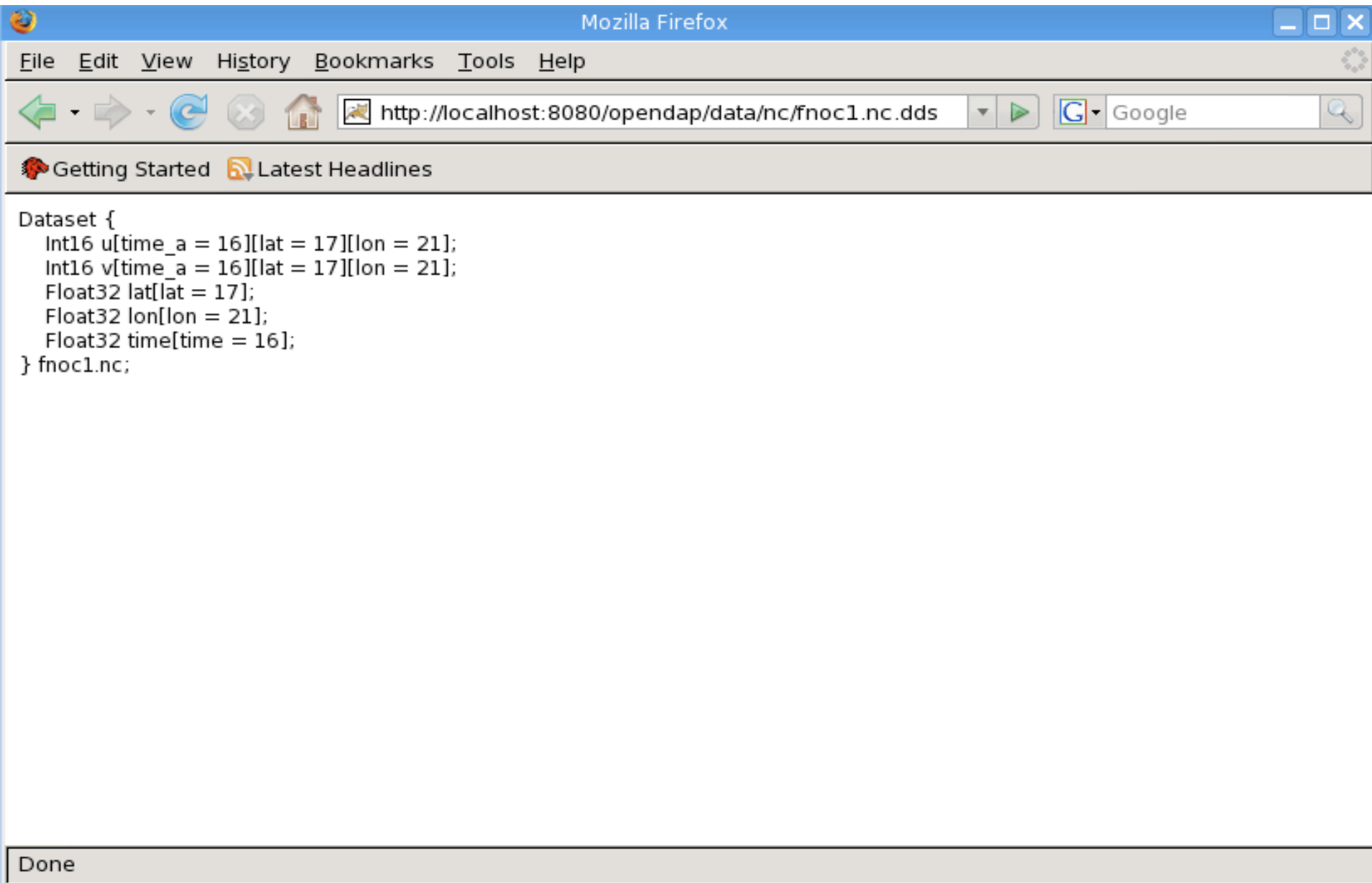
THREDDS Catalog [HTML](#) [XML](#)

OPeNDAP Hyrax (1.2.1) [Documentation](#)

Done

Hyrax development sponsored by [NSF](#), [NASA](#) and [NOAA](#)

... and get the DDS



How to write format files for FreeForm

FreeForm

- Provides a way to serve data stored as ASCII, binary and Dbase files
- Data are organized in rows where each row is a record
- Within each record, the fields are denoted using column positions (like old FORTRAN Format statements)
- For each data file, there must be a matching format file
- Each row must be exactly the same
 - Use padding for string data of varying lengths

FreeForm, cont.

- Data files may have headers and those headers need not follow the same layout as the data rows which follow
- The information shows up as global attributes in the DAS/DDX responses

How to write a FF format file

Each file has an input and an output section; the DAP handler uses only the input, but the output is useful if you need to use FreeForm's diagnostic tools.

```
ASCII_input_data "URI_Avhrr"
```

```
year      1 4      int32 0
```

```
month     6      int32 0
```

```
day       9 10     int32 0
```

```
hours     12 13     int32 0
```

```
minutes   15 16     int32 0
```

```
seconds   18 19     int32 0
```

```
DODS_URL  21 93     text  0
```

Field name, starting column, ending column, data type value scale factor

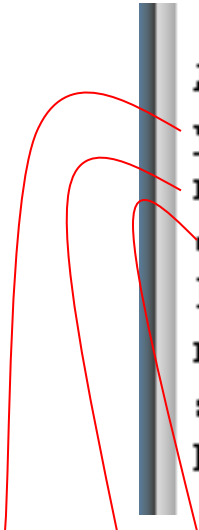
```
// The output_data part of the file is not
```

```
// However, tools like newform use it. 8/13
```

```
ASCII_output_data      "ASCII_data"
```

```
year      1 4      int32 0
```

Compare format to data



```
ASCII_input_data "URI_Avhrr"
year      1 4      int32 0
month     6 7      int32 0
day       9 10     int32 0
hours    12 13     int32 0
minutes   15 16     int32 0
seconds   18 19     int32 0
DODS_URL 21 93     text   0
```

1979/04/11:18:53:59 http://dods.gso.uri.edu/cgi-1185359.pvu.Z

1979/04/15:19:48:52 http://dods.gso.uri.edu/cgi-5194852.pvu.Z

1979/04/16:19:40:24 http://dods.gso.uri.edu/cgi-6194024.pvu.Z

1979/04/17:08:00:11 http://dods.gso.uri.edu/cgi-7080011.pvu.Z

1979/04/17:09:49:59 http://dods.gso.uri.edu/cgi-7080011.pvu.Z

More FreeForm features

- Headers
 - File headers: Use “ASCII_file_header <title>” in the format file to denote the format of a file header
 - Record headers are also supported
 - Headers must be a single record
- Additional attributes
 - A file named <data>.das will be used as a set of ancillary attributes
 - Those will be merged into the attributes returned by the handler

Examples

- See the FreeForm sample data in /usr/local/share/hyrax/data/ff
 - ‘.dat’ denotes an ASCII data file
 - ‘.fmt’ is the format file
 - ‘.das’ is the extra attribute information